

# Lab 1 — First Commit · Tonight in 20 steps

OPIM 5512 · Module 1  
Partner A = weather · Partner B = demand

**The deliverable tonight is a repo, not a notebook.** Two halves, one merge. **No AI tonight.** Keep the instructions open beside Colab — every click is in there.

## Set up steps 1–6

- 1 **Pair up.** Decide who is Partner A (weather) and Partner B (demand). Solo/online? Two accounts, or pair over Teams.
- 2 **Partner A:** create a new **public** repo for the pair.
- 3 **Partner A:** add Partner B as a **collaborator** → Partner B **accepts** the invite.
- 4 **Partner A:** turn on **branch protection** for `main` (require a PR + 1 approval).
- 5 **Both:** **clone** the repo **once** in GitHub Desktop.
- 6 **Each:** make your branch ( `dev-weather` / `dev-demand` ) → **Publish** it.

## Grab data & build your half steps 7–13

- 7 **Each:** open your **starter** in Colab (new tab) → run the first cell and **look at your raw data**.
- 8 **Together:** write the **data dictionary** (your contract) in `README.md` — names, units, what a timestamp means.
- 9 **Each:** clean your half in Colab → **File** → **Save** to your branch (drop a `#TEST` to prove it saved).
- 10 **Each:** save your **CSV** and your **plot**; download both.
- 11 **Each:** GitHub Desktop → **Show in Explorer** → drag CSV → `data/clean/`, plot → `images/`.
- 12 **Each:** **commit** (real message) → **push**.
- 13 **Each:** open **one pull request** (your branch → `main`): notebook + contract + CSV + plot.

## Review & merge steps 14–17

- 14 **Each:** request your **partner** as reviewer.
- 15 **Each:** **review your partner's PR** — read the diff, then **Approve**.
- 16 **Merge** both PRs → **delete** the branches. Whoever merges the `README` **second** resolves the **merge conflict** — keep both halves.
- 17 **Both:** switch to `main` → **Fetch** → **Pull**. Now you each have **both** files.

## Join & report steps 18–20

- 18 **Together (one screen):** open the **joint EDA** notebook → run the **join** → find **three things**, one surprising.
- 19 **Write** `REPORT.md` (from the template) + save figures → branch → PR → partner reviews → merge.
- 20 **Read-out:** one figure on screen, one sentence — check your **network graph** shows the loop both ways.

### Don't panic

- **Commit early** — committed work is nearly impossible to lose.
- **Fetch & Pull can't hurt you.** Fetch = peek, Pull = download.
- **Same path every save**, or you make a duplicate notebook.
- **Clear all outputs** before a notebook goes in.
- `main` rejecting your push = the protection working.

### Done at 7:30 when the repo has

- Branch protection on `main`; both partners collaborators
- `README.md` data dictionary for **both** files
- `data/clean/weather_hourly.csv` + `demand_hourly.csv`
- Both notebooks + joint EDA · `REPORT.md` + figures
- **≥3 merged PRs**, branches deleted, both authoring **and** reviewing
- A network graph showing the loop both ways